Covariant Functors of Directed Containers

Extracted from a machine-checked development

Abstract

We introduce covariant functors between directed containers: maps that act forward on both shapes and positions and preserve root, sub-shape, and shift. Under the dictionary that a directed container presents a small category with morphisms $p : s \to s \downarrow p$, such a functor is precisely a functor of presented categories. We construct identities and composition, prove the category laws, and record the transport lemmas needed for the composition law on shifts. The proofs of all stated results have been machine-checked; we record only the mathematical content.

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1 Introduction

A directed container presents, for each shape s, a category whose morphisms out of s are positions of s, with identity the root and composition given by the shift. Container morphisms in the classical sense are *contravariant* on positions and correspond to cofunctors of these presented categories. The present note isolates the covariant sibling: a map that sends shapes forward and positions forward, preserving root, sub, and shift.

The definition is given directly on directed containers, without passing through an intermediate "small category" construction. This keeps the identity and unit laws definitional on containers whose sub-shape operation is strictly a projection, and matches the law-naming style of the corresponding cofunctor development.

We work in a single universe of types. Proofs are indicated by the phrase "by lean."

What this note claims. Directed containers and their covariant functors form a category. A covariant functor $C \Rightarrow D$ is a shape map and a forward position map satisfying three laws F0–F2 (preservation of sub-shapes, roots, and shifts). Identities and composition exist, composition is associative, and identities are strict units.

2 Background: directed containers

Definition 2.1 (Directed container)**.** A **directed container** consists of a container (S, P) together with

$$\begin{aligned} \mathrm{o} &: (s : S) \to P(s), \\ \downarrow &: (s : S) \to P(s) \to S, \\ \oplus &: (s : S) \to (p : P(s)) \to P(s \downarrow p) \to P(s), \end{aligned}$$

satisfying the standard axioms D1–D5.

Remark 2.2 (Presented category)**.** Informally, one may regard each shape s as an object and each position $p \in P(s)$ as a morphism $s \to s \downarrow p$, with $\mathrm{id}_s = \mathrm{o}(s)$ and composition $(p; q) = p \oplus q$. The axioms D1–D5 are the category laws for this presentation. We use this dictionary only as motivation; the formal development never builds the category explicitly.

3 Covariant functors

Definition 3.1 (Covariant functor of directed containers)**.** Let C and D be directed containers. A **covariant functor** $\varphi : C \Rightarrow D$ consists of

1. a map on shapes $u : \mathrm{Shape}(C) \to \mathrm{Shape}(D)$, and
2. for each shape $s \in \mathrm{Shape}(C)$, a map on positions

$$g_s \;:\; \mathrm{Pos}(C)(s) \;\longrightarrow\; \mathrm{Pos}(D)(u(s)),$$

such that the following three laws hold:

$$u(s \downarrow_C p) \;=\; u(s) \downarrow_D g_s(p), \tag{F0}$$

$$g_s(\mathrm{o}_C(s)) \;=\; \mathrm{o}_D(u(s)), \tag{F1}$$

$$g_s(p \oplus_C q) \;=\; g_s(p) \oplus_D \big((\mathrm{F0}) \triangleright g_{s \downarrow_C p}(q)\big). \tag{F2}$$

Here the transport in F2 moves the inner image position $g_{s\downarrow_C p}(q)$, which lives over $u(s \downarrow_C p)$, into the fibre over $u(s) \downarrow_D g_s(p)$ along the equality F0.

Remark 3.2. Contrast classical directed-container morphisms (cofunctors of the presented categories), whose position component is *contravariant*. Here both components run forward: F0 says the codomain of a morphism is preserved, F1 preserves identities, and F2 preserves composition.

4 Identity and transport lemmas

Definition 4.1 (Identity functor)**.** For a directed container C, the **identity functor** $\mathrm{id}_C : C \Rightarrow C$ has shape map and position map both the identity. The laws F0–F2 hold by reflexivity; the transport in F2 is along a reflexive equality and vanishes.

Lemma 4.2 (Position map commutes with transport)**.** *Let $\varphi : C \Rightarrow D$ be a covariant functor, let $e : a = b$ be an equality of shapes of C, and let p be a position of a. Then*

$$g_b(e \triangleright p) \;=\; (u(e)) \triangleright g_a(p), \tag{1}$$

where $u(e)$ denotes the image of e under the shape map.

Proof. By lean. □

Lemma 4.3 (Composite transport)**.** *For a family P and equalities $h_1 : x = y$, $h_2 : y = z$,*

$$h_2 \triangleright (h_1 \triangleright p) \;=\; (h_1 \,;\, h_2) \triangleright p. \tag{2}$$

Proof. By lean. □

5 Composition

Definition 5.1 (Composition of covariant functors)**.** Let $\varphi : C \Rightarrow D$ and $\psi : D \Rightarrow E$ be covariant functors, with shape maps u, u' and position maps g, g' respectively. Their **composite** $\psi \circ \varphi : C \Rightarrow E$ has

$$(\psi \circ \varphi)_S(s) \;=\; u'(u(s)), \tag{3}$$

$$(\psi \circ \varphi)_p(s, p) \;=\; g'_{u(s)}(g_s(p)). \tag{4}$$

The law F0 for the composite is the composite of the factors' F0 equalities; F1 is two rewrites along the factors' F1; F2 is the factors' F2 followed by Lemmas 4.2 and 4.3.

Proposition 5.2 (Left unit)**.** *For every covariant functor $\varphi : C \Rightarrow D$,*

$$\varphi \circ \mathrm{id}_C \;=\; \varphi. \tag{5}$$

Proof. By lean. □

Proposition 5.3 (Right unit)**.** *For every covariant functor $\varphi : C \Rightarrow D$,*

$$\mathrm{id}_D \circ \varphi \;=\; \varphi. \tag{6}$$

Proof. By lean. □

Proposition 5.4 (Associativity)**.** *For covariant functors $\varphi : B \Rightarrow C$, $\psi : C \Rightarrow D$, and $\chi : D \Rightarrow E$,*

$$\chi \circ (\psi \circ \varphi) \;=\; (\chi \circ \psi) \circ \varphi. \tag{7}$$

Proof. By lean. □

Remark 5.5. The data of functors compose strictly. The law fields are propositional, so associativity and the unit laws hold by proof irrelevance once the underlying maps agree.

6 Discussion

What has been established. Covariant functors of directed containers form a category: they have identities, an associative composition, and strict unit laws. The only genuinely dependent obligation is F2, which requires transporting an inner position along F0 before the target shift can consume it; composition of functors inherits this obligation and discharges it via the two transport lemmas.

Relation to cofunctors. The classical morphisms of directed containers (position-contravariant, shape covariant) are the cofunctors of the presented categories. The structure defined here is the functorial dual. Both live over the same objects; they are not interchangeable.

Scope. The development does not construct the presented small category as an independent object, nor does it prove an equivalence between covariant functors of directed containers and functors of those categories. It also does not treat natural transformations between covariant functors.

Machine checking. Every definition and theorem statement above corresponds to a declaration in a machine-checked formalisation. The phrase "by lean" indicates that the proof has been verified by a proof assistant.

Acknowledgements

The mathematical content is a covariant counterpart to the cofunctor calculus for directed containers of Ahman, Chapman, and Uustalu. The presentation here is a prose rendering of a formal development intended for human review of the theorem statements.

References

[1] D. Ahman, J. Chapman, and T. Uustalu. When is a container a comonad? *Logical Methods in Computer Science*, 10(3), 2014.

[2] M. Abbott, T. Altenkirch, and N. Ghani. Containers: constructing strictly positive types. *Theoretical Computer Science*, 342(1):3–27, 2005.